

Software Engineering

How To Create Software Products Everyone Loves!

Module 9

UI/UX Fundamentals: Typography

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Font Structure

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Kerning and Whitespace

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Fonts and Design Hierarchy

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Readability, Font Types and Mobile Devices

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Font Paring Rules

Typography

The fonts you choose profoundly influence the look & feel, style, and message of the entire project

Aa Aa Ad Aa Aa

FONT OR TYPEFACE?

A **Typeface** is a collection of various styles of the same lettering. Each of those styles is a **font**.

Helvetica is a typeface, and Helvetica Bold is a font.

HOW TO CHOOSE THE RIGHT FONT?

If the product you're designing is going to be multi-lingual, check if your font supports all the letters like **Special Character.**

Aa Ee
Óó Ćć

HOW TO CHOOSE THE RIGHT FONT?

Choose a font that has a **variety of weights**. It will help you set the right hierarchy between elements

Aa Aa

Aa Aa

HOW TO CHOOSE THE RIGHT FONT?

Simple is good.

Choose a font that's
not overly
ornamental or
fancy. The main
goal should always
be readability.

Ad Aa

Aa Ad

HOW TO CHOOSE THE RIGHT FONT?

Check a font in all of its sizes for the sake of **readability**. If you can still read it at a tiny size and it looks good while super-big, you should be safe.

Aa Aa

Aa Aa

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FONT STRUCTURE

Each font has a set of characteristics, based on its structural definition.



BASELINE

The baseline is a flat surface on which our font is sitting. You can use it to align the type to other on-screen elements.

The word "Type" is displayed in a dark blue, serif typeface. A horizontal light blue line, representing the baseline, runs through the bottom of the letters. The letters 'y' and 'p' extend below this line, while 'T', 'e', and the second 'p' sit on or above it.

CAP-HEIGHT

Cap height is the height of the capital letter.



CType

X-HEIGHT

The height of lowercase x is as a default value for other, less uniform lowercase letters.



ASCENDERS

All elements above our x-height are called ascenders.

The word "Type" is displayed in a dark blue serif font. A light blue horizontal bar highlights the top of the letter 'T', which is an ascender. The bar extends across the width of the 'T' and slightly beyond its left and right edges. The letters 'y', 'p', and 'e' are positioned below the baseline of the 'T'.

DESCENDERS

All elements below
our x-height are
called descenders



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KERNING

Kerning is the space between letters. All fonts have a default kerning value that differs between them.



The image shows the words "Typography" and "Wok" in a dark blue, sans-serif font. Vertical light blue lines are placed between the letters of "Typography" to show the space (kerning) between them. There are nine such lines, one between each pair of adjacent letters. For the word "Wok", there is a light blue line between the 'o' and 'k', and a diagonal light blue line between the 'W' and 'o'.

KERNING

When the kerning is too small, or too large, it can negatively impact readability.

Typography



Kerning +1
Too wide

Typography



Kerning 0 (default)
Readable

Typography



Kerning -1
Still Readable

Typography



Kerning -2
Too narrow

KERNING

Leading is the vertical space between lines of text.

20 [Typography is one of the broader subjects in design,
12 [and there are many books dedicated to this matter.

WHITESPACE

Whitespace is the empty area between letters, words, lines of text, and everything else we see on-screen.

Too small or too large whitespace value negatively impacts readability and hierarchy.

Whitespace is essential to a good layout, and getting it right takes a lot of practice

SAFE WHITESPACE

This example shows large enough margins and spacing for all the elements to be readable.

Typography

Introduction

Typography is one of the broader subjects in design, and there are many books dedicated to this matter.

TOO SMALL WHITESPACE

When the kerning is a very low number, and everything is very close together, it's a lot harder to read and process.

O Typografiii

Introduction

Typography is one of the broader subjects in design, and there are many books dedicated to this matter.

TOO LARGE
WHITESPACE

Uneven margins with very high whitespace make design look chaotic and hard to read. It seems as if there are no rules applied to the layout.

Typography

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Font Paring Rules

HIERARCHY AND GRADATION

Gradation helps with building hierarchy by adding more visual impact to top-level, essential elements like headings and titles.

You can use both the size and weight of your font to create gradation and visual hierarchy

GRADATION AND HIERARCHY

This content has a large heading, slightly smaller subheading, and much smaller body copy. That gradation defines the hierarchy and makes it much easier to understand the purpose of every element.



NO GRADATION

Without gradation, there's no hierarchy. When every element has the same visual weight, it makes everything look the same. You don't know where the title ends, and the copy begins.



HOW TO SET THE HIERARCHY

The Golden Ratio (1.618) has many uses in design.

Good practices

The easy way out.

Typography is one of the broader subjects in design, and there are many books dedicated to this matter.

H2 16p

H1 26p

P 10p

GRID

If you add a horizontal grid, you can use it to align text and create text rhythm.

4x
4x
2x

Typography

11.11.2019

Typography is one of the broader subjects in design, and there are many books dedicated to this matter.

Typography

11.11.2019

Typography is one of the broader subjects in design, and there are many books dedicated to this matter.

ALIGNMENT

Text alignment is a cultural thing and can differ between countries.

Try **to avoid justified text** while designing digital products. Having both text edges uniform may seem friendly **but it happens at the cost of irregular kerning** and varying spaces between individual words

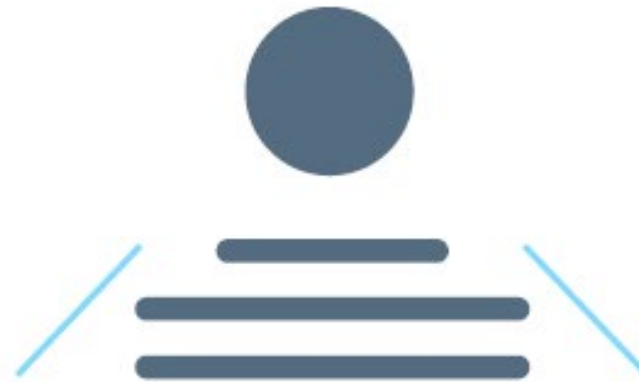


CENTERED TEXT

Center aligned text works well when it's short. You can use center-align for two or three sentence blocks. The best shape of the text is a triangle



Triangle shape



Triangle shape



Irregular shape

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LINE LENGTH

If you're designing text blocks for mobile devices, try to keep within **30 to 50 characters per line.**

On larger screens (like tablets and laptops), the best range is between 6 and 9 words per individual line.

CONTRAST

Very **high contrast** combined with **shining light** and a **close-range** can be very unpleasant.

While you should always think about having an **accessible** design, try not to overdo it with ultra-high contrasts either.

CONTRAST

While designing text blocks, try to avoid pure-black (#000000). It's much better to use dark grey (like #222222) or better yet use a grey that's mixed with your primary color.

Typography

Introduction

Typography is one of the broader subjects...

PRIMARY ACTION

The primary dark blue CTA defines a blue-grey shade for our fonts. That makes every part of the info card internally consistent and more pleasant to the eye.

SANS-SERIF FONTS

Sans-serifs are well suited to digital products because of their simplicity in shape and form. They look good even on lower-quality displays.

a Aa Ad Ad A

WHEN TO USE SANS-SERIF FONTS?

They work especially well for both headings and mid-length text blocks, descriptions, labels, buttons, and forms.

Sans-serifs are the base font style for all digital interfaces.

SERIF FONTS

Serif fonts got their name from having little additional lines (called serifs) at the character edges. They also used to be called Roman fonts.

Aa Aa Aa Aa Aa

WHEN TO USE SANS-SERIF FONTS?

Serif fonts are great for print media (books, newspapers, and magazines).

In digital products, they should co-exist with sans-serif fonts. Using only serifs in an app or on a website is not recommended unless the choice is branding related.

DECORATIVE FONTS

Decorative fonts are script, calligraphy, handwritten, and all the other stylized typefaces.



The image displays four pairs of letters, 'Aa', in different decorative font styles. From left to right: 1. A dark blue script font with elegant, flowing curves. 2. A dark blue modern sans-serif font with clean, straight lines. 3. A dark blue classic serif font with traditional flourishes. 4. A dark blue bold, rounded sans-serif font with a friendly, approachable feel.

FONTS TO AVOID

Some fonts are simply not made for UI and high-density displays

Arial

Times New Roman

Tahoma

Courier

Verdana

Calibri

Comic Sans

Trebuchet

MOBILE DEVICES AND FONTS

Both Android and iOS have their dedicated fonts that should be your first choice for designing apps on these platforms.

Try to use mobile system fonts in your first designs exclusively. **The users already know** those typefaces from all the other apps they are using, **so your design will instantly seem familiar.**

MOBILE DEVICES AND FONTS

Try not to mix the
fonts between the
platforms

San Francisco (SF UI)

Apple (iOS, Mac OS, Watch OS, TV OS)

Roboto

Android, Material Design (Mobile + Web)

FONT SIZES ON MOBILE

Both Apple and Google also suggest the smallest readable font-size for body-copy and inputs.



FONT WEIGHT (LIGHT, ULTRA LIGHT)

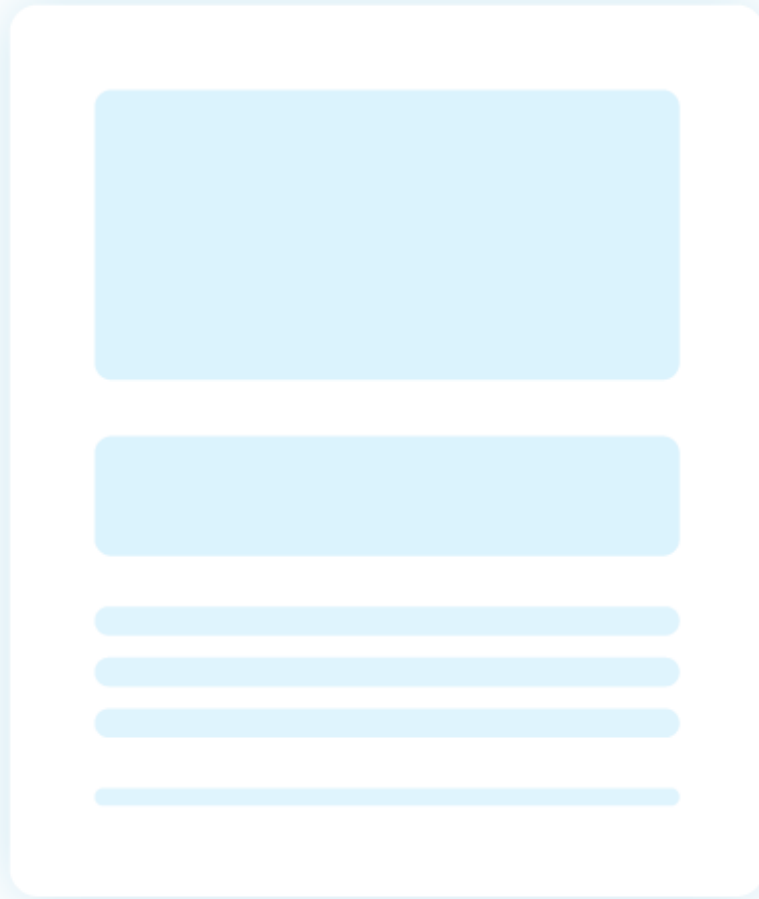
For a thin font to be readable, it has to be the right size.

This size is enough for that weight.

This is too thin.

This will be super hard to read.

CSS (Cascading Style Sheets)



H1

H1 is the main title.

H2

H2 is a subtitle.

P

P is paragraph text or body copy
- used for longer blocks of content.

=

Span

Span is often a way to specify
all meta-data like publication
date, comment count, author
name, etc.

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Font Paring Rules

FONT PAIRING RULES(FONT WEIGHT)

This is an optimal
ratio of heading to
paragraph font
weights



Pairing fonts

The basics of font pairing
are quite simple.

FONT PAIRING RULES(FONT WEIGHT)

The difference in weight can't be too extensive. In this example, the header is visually heavy and makes the paragraph text disappear.



Pairing fonts

The basics of font pairing are quite simple.

FONT PAIRING RULES(FONT WEIGHT)

Headings should nearly always use a ticker font than the paragraph text. In this example, the hierarchy is hard to understand.



Pairing fonts

**The basics of font pairing
are quite simple.**

FONT PAIRING RULES(TWO SANS-SERIFS)

Two distinct sans-serif typefaces can work well together. The secret is choosing ones that are different enough from each other.



Pairing fonts

The basics of font pairing are quite simple.

Montserrat + Work Sans

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

While trying to pair a serif typeface with a sans-serif one, try to choose the simplest possible sans-serif font. The serifs are usually complicated enough visually.



Pairing fonts

The basics of font pairing are quite simple.

Playfair Display + Work Sans

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

A more decorative serif font combined with a decorative sans-serif makes the text block complicated and hard to read

Pairing fonts

The basics of font pairing are quite simple.

Playfair Display + Brandon Grotesque

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

If you plan to use a serif font for paragraph text, try matching it with a thicker sans-serif heading for best results.



Pairing fonts

The basics of font pairing are quite simple.

Montserrat + Georgia

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

Choosing a thinner sans-serif can severely impact the readability of the entire text-block.

Pairing fonts

The basics of font pairing are quite simple.

Montserrat + Georgia

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

Entirely serif-based designs should be avoided unless that choice is brand-related.

Pairing fonts

The basics of font pairing are quite simple.

Georgia

FONT PAIRING RULES(SERIF AND SANS-SERIF MIX)

Two different serif fonts are also hard to do right. In most cases, the result will be too complicated visually.

Pairing fonts

The basics of font pairing are quite simple.

Playfair Display + Georgia

THE RIGHT NUMBER OF FONTS

When it comes to the number of typefaces, weights, and sizes, try to keep it simple.

If it's possible, try to choose one or two fonts, with two weights (regular and bold) and between 3 and 5 sizes.

The next step is to use those rules consistently across your design.

THE RIGHT NUMBER OF FONTS

This example uses
Work Sans with
three weights (bold,
medium, regular)
and size gradation
of 24,16,12 and 8.

Heading (h1)

Work Sans Bold (24px)

Subheading (h2)

Work Sans Medium (16px)

Paragraph text / body copy (p).

Work Sans Regular (12px)

Smallest text, used for meta-data and labels (span).

Work Sans Regular (8px)

Typographic consistency

Typographic consistency is essential for all the interactive parts of the interface, such as buttons and forms.

PRIMARY BUTTON

Work Sans Medium (8px)

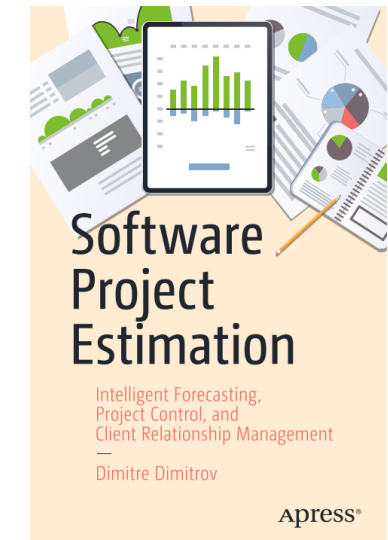
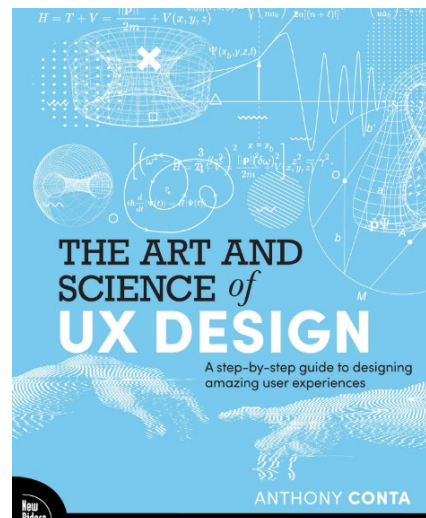
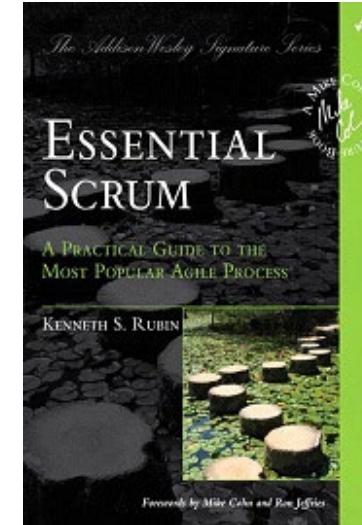
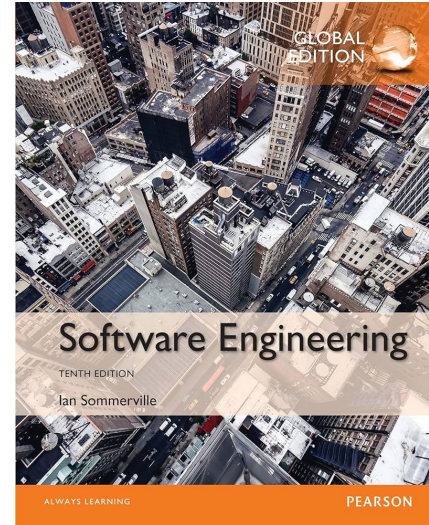
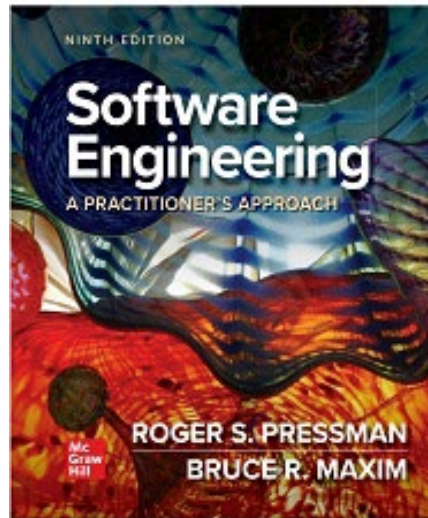
SECONDARY BUTTON

Work Sans Medium (8px)

GHOST BUTTON

Work Sans Medium (8px)

Course References



Course References

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